Specify a Linux Distribution

- Hardware platform.
  - Section A - HLX Hardware Overview Document
  - Section B - 00-444633 LX SCALE MAIN BOARD Schematic
  - Section C - 00-445088 CONSOLE CONTROL 5-WIRE Schematic
  - Section D - Embedded Software Group USB Product Id Table
  - Section E - Access Scale PPT presentation
  - Section F - HLX PPT presentation
  - Section G - LX Service Scale presentation
- List of software modules/middleware outside those supplied with the current OS
  - Section H- HLX Version Info
- List of software modules/middleware supplied/built into application with current OS
  - Section I - Module build report
- Memory Map
  - Section J – Memory and I/O map
- General Requirements
  - Must be a distribution which has long term support until the end of 2013
  - Must be from acknowledge source of Linux distributions
  - Must support a major long term supported kernel release. As referenced through the Linux Foundation
  - Must support General Software (CalSoft) version 1.1 BIOS for the Geode LX800.
  - Must include any I/O drivers as either part of the kernel or a supported package readily available from the distribution vendor or the device vendor.
  - Must support at a minimum the ex3/ex4 and FAT32 file systems. Must be able to partition devices, Compact Flash, Thumb drives, Hard Disks for multiple file system support.

Section A - HLX Hardware Overview Document

HLX Hardware Overview Document

filename: HLX HW overview.doc

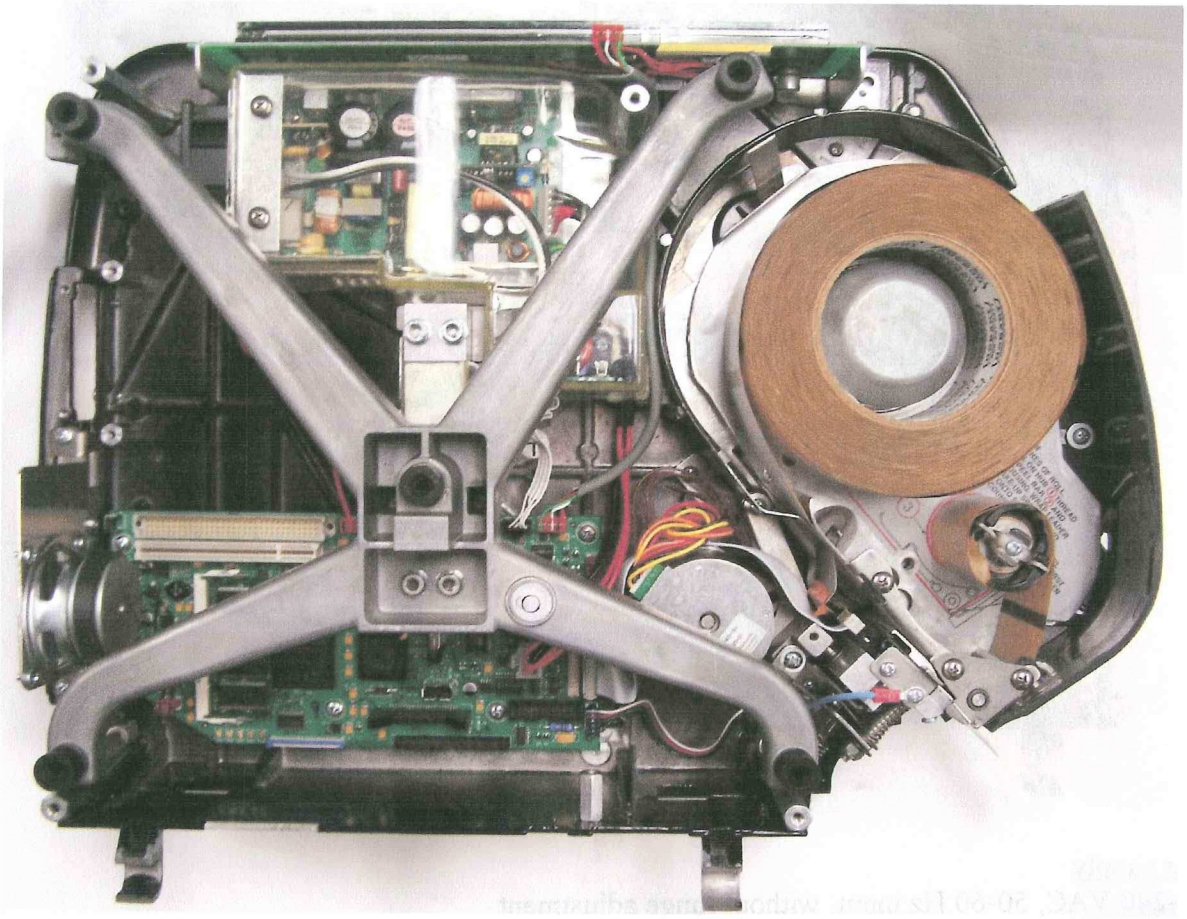
Revision History

4/23/2008	MEE	Created
4/24/2008	MEE	Updated

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HLX Scale/Printer Base (with load cell & spider)



Weigher

- Load cell – HBM PW2R, 15kg, 5000d accuracy
- Dual range – 30 lb * .01 & 12 lb * .005 (or 15 kg * 5g & 6kg * 2g), tested for NTEP & OIML
- Most difficult test to pass is "return to zero" on creep test for dual range
- Overload stops straddle load cell, corner stops transmit load from spider to base
- ESD path from platter through conductive rubber pads, to spider, load cell & base

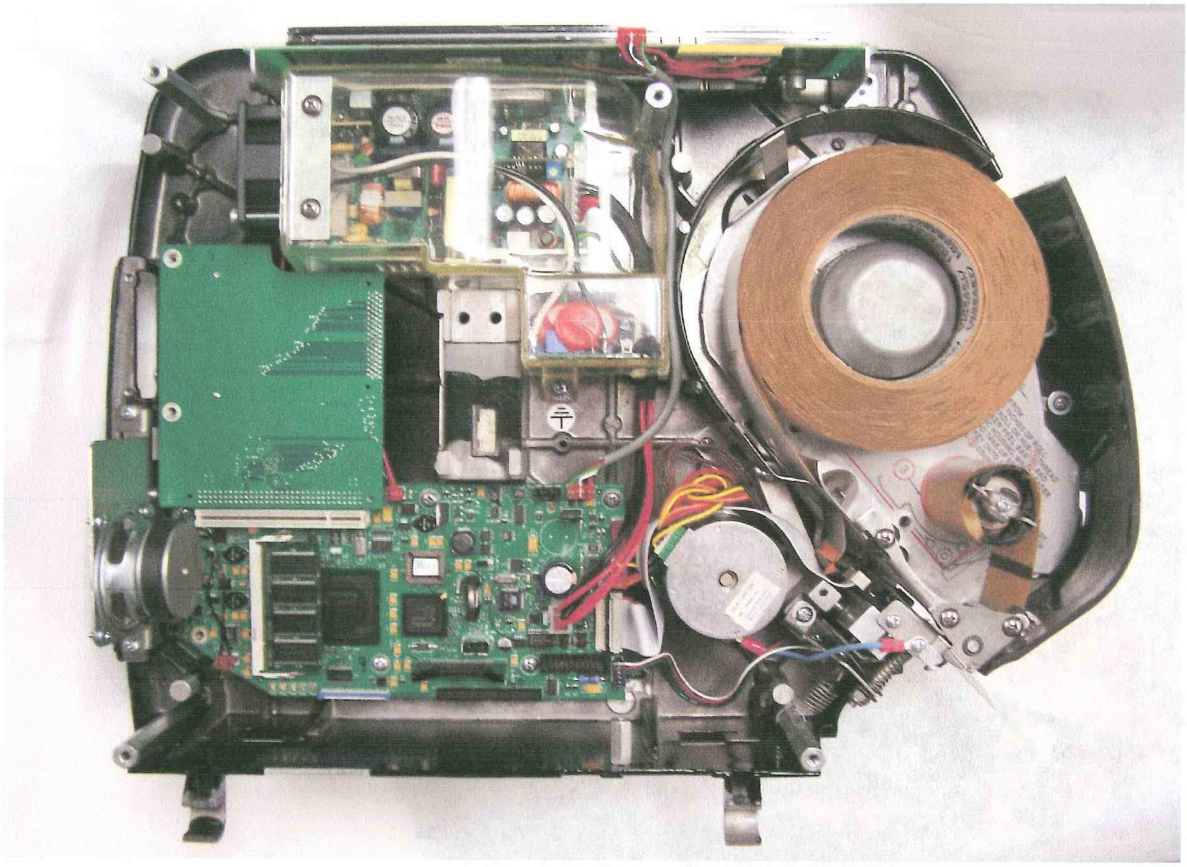
Printer

- Printhead – TDK BHF4251, 448 dot, 8 dot per mm, thin film, flat glaze type
- Stepper motor – half stepped at 16 per mm to reduce drive train noise
- Media sync sensor – retroreflective type, monitors black bars printed on backing paper, includes head up & cassette type sensors

Main board

- Fastened to base via 6 screws
- Can be removed without disturbing load cell/spider or speaker

HLX Scale/Printer Base (with Cardbus adapter)

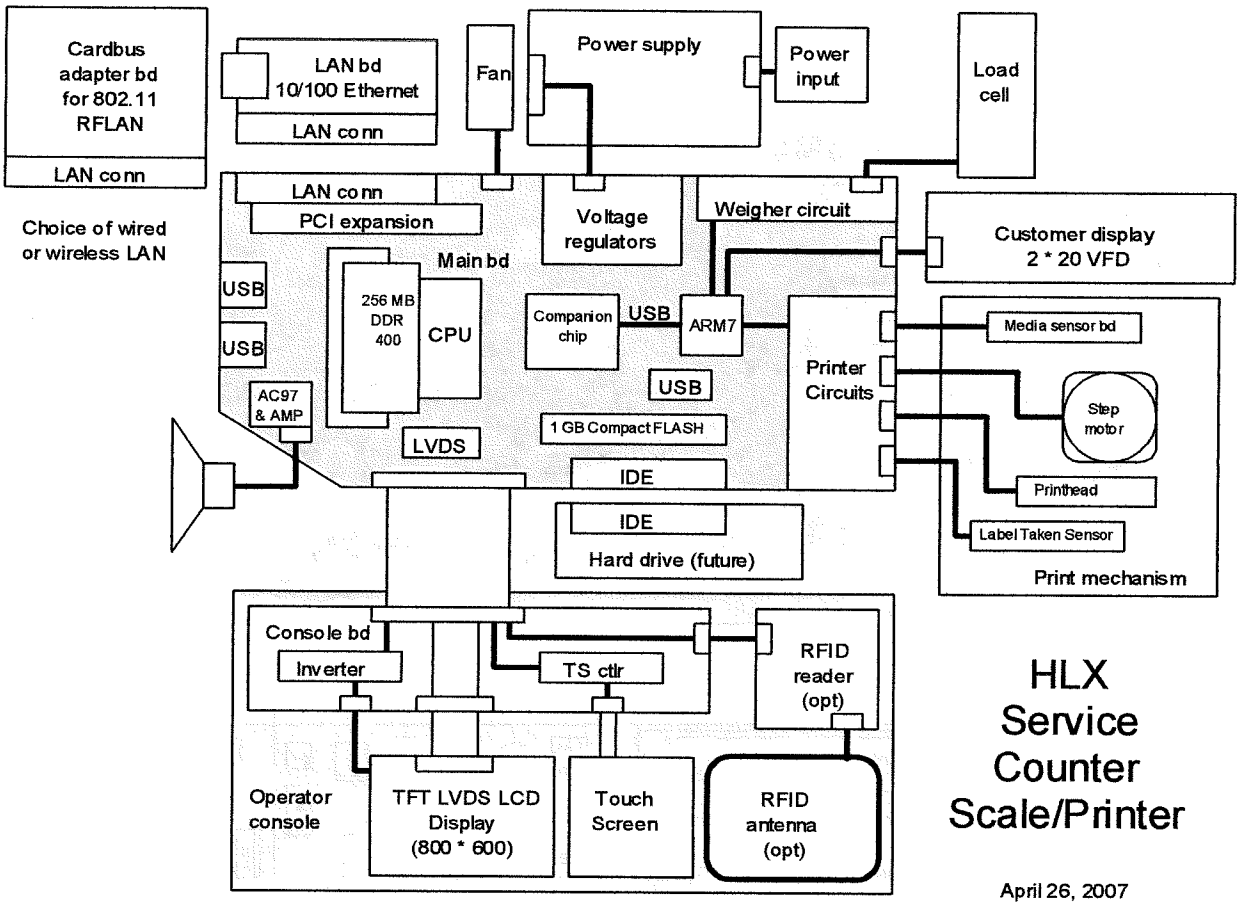


Power supply
- 100-240 VAC, 50-60 Hz input, without range adjustment
- 24 VDC 150 W continuous output, 200W peak
- Multiple sourced – Amperor, Power One, APS, Hitek
- Fan – 2 speed, air intake at left side of base (& other areas),
- Cover directs airflow over power supply & exhaust through base
- Power input board – fuse & surge suppression

Speaker
- 2W rated, 2W RMS drive, Mylar cone, sound exits left side of base

Customer display
- 2 line by 20 large character vacuum fluorescent
- Purchased custom module from Futaba

HLX Block Diagram



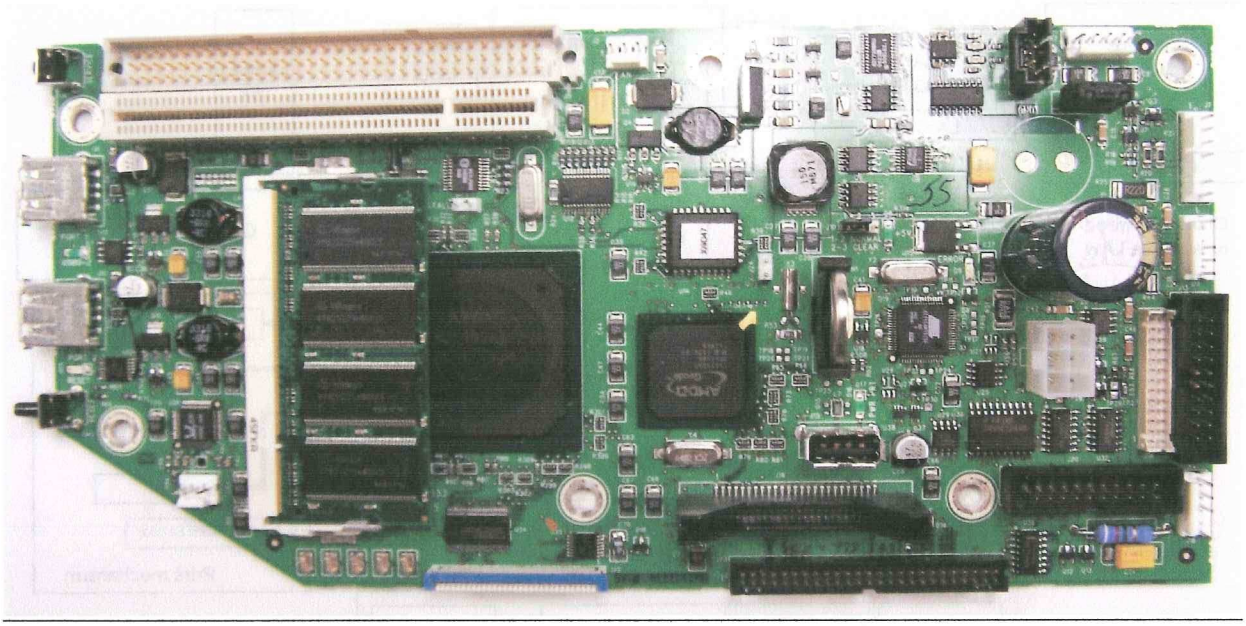
- Power supply connects to main bd which feeds all other components

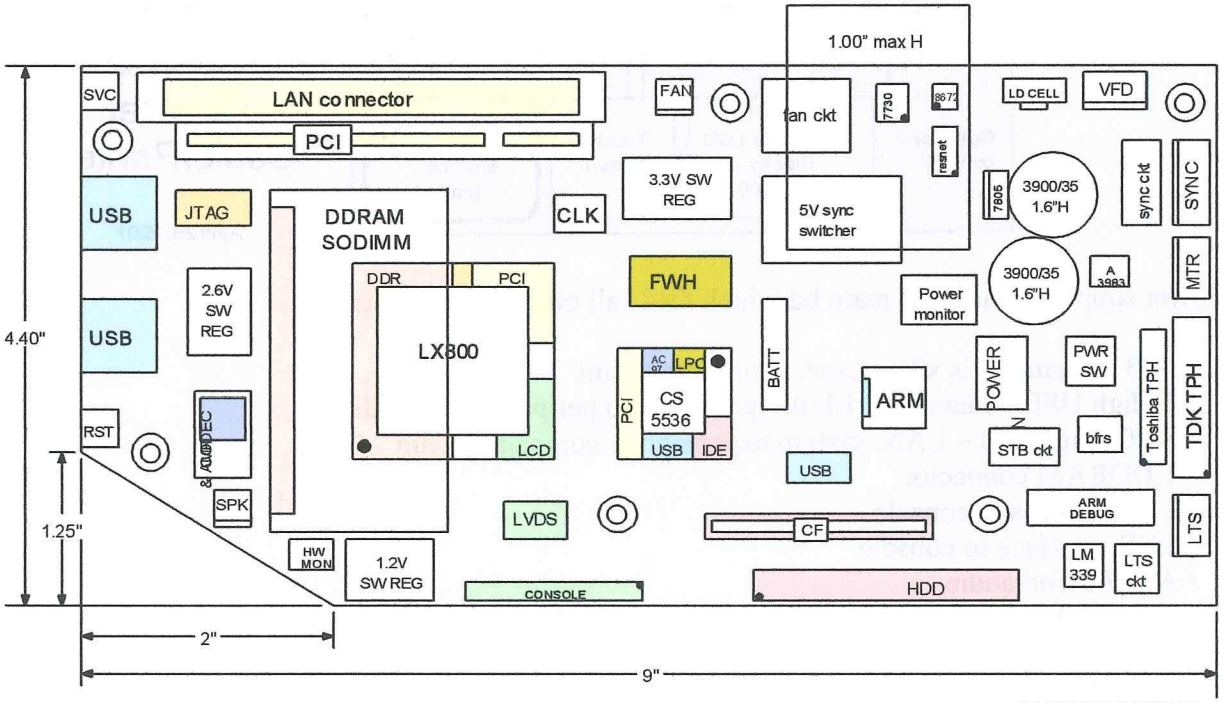
- Left 2/3 of main bd is x86 system controller circuit
  - 3 High USB connectors; 1 Full Speed port to peripheral controller
  - 3 PCI functions – LAN, system expansion & companion chip
  - 1 DDRAM connector
  - 2 COM ports to console
  - LCD interface to console
  - AC97 mono audio
  - Parallel ATA ports for Compact FLASH & Hard Disk Drive

- Right 1/3 of main bd is ARM7 peripheral controller circuit
  - Weigher circuit interfaces to load cell
  - UART interfaces to customer display
  - Various I/O (A/D, SPI, timers, etc) drive printer mechanism

- Console has LVDS LCD display, 5 wire touch screen, console board & optional RFID circuit

Main bd photo & layout



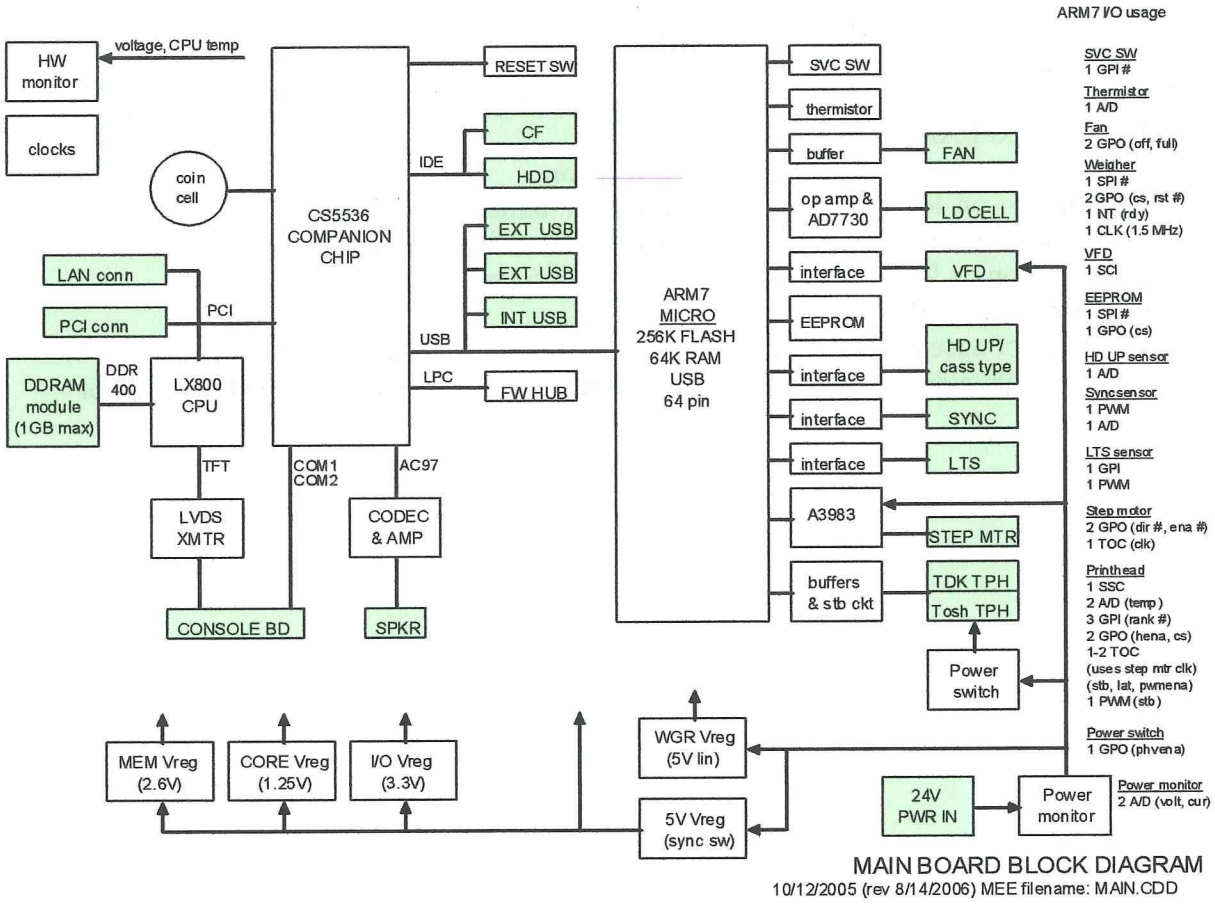


MAIN BD LAYOUT
2/26/2006 MEE (rev 7/10/2006)
file: MAINBD3.CDD

•PC bd is 8 layer, high speed bus design (PCI, DDR, USB, LVDS)
• Passive parts on back side
• No heat sink on x86 CPU (AMD LX800)

- RAM module is standard 512 MB DDR400, notebook style, 1 GB max
- 3 High speed USB ports (2 external, 1 internal)
    - Typical peripherals are keyboard, thumb drive
    - Have used USB 802.11b/g RFLAN
- LAN connector with PCI signaling dedicated to either 10/100 Ethernet bd or Cardbus adapter bd (for 802.11b/g RFLAN cards)
- PCI connector (standard pinout) for system expansion
- Reset pushbutton on left side behaves like Power button on PC; press to turn on, press to perform Windows shutdown, hold for 4 seconds for immediate shutdown
- Service pushbutton on left side accesses weigher calibration functions
- Choice of Compact FLASH or hard drive
- Console connector accepts 30 conductor flat flex cable for pivoting console
- Connectors on right side for printhead, stepper motor, label sync sensor, label taken sensor

Main Board Block Diagram



MAIN BOARD BLOCK DIAGRAM

10/12/2005 (rev 8/14/2006) MEE filename: MAIN.CDD

System processor

- 500MHz AMD LX800 x86 processor performs like 800 MHz but at 1.6W typical
- DDRAM, LCD & PCI interfaces located on LX800 IC
- Single DDRAM module needs only series termination
- TFT signals also series terminated to reduce radiated noise
- Other unused functions – video input, analog video out, security (AES, RNG, etc)

Chipset

- Companion chip is AMD CS5536, connected via PCI to LX800
- CS5536 has 4 High Speed USB, 2 COM, IDE, LPC, AC97 audio ports
- Real Time Clock built in to CS5536
- Contains ACPI circuits using RESET switch and shutdown command for orderly Windows power up & shutdown

FW Hub (NOR FLASH)

- Firmware hub is Low Pin Count style, soldered to main bd, contains General Software BIOS
- Can boot from Compact FLASH, Hard drive, USB thumb drive
- Access to BIOS via USB keyboard

Audio
• CODEC & monaural Class D amplifier drive single speaker

Console interface
• Standard TFT signals converted to LVDS for least EMI, best signal quality, lower pin count
• COM1 & COM2 channels are buffered & used for touch screen & RFID
• 30 pin vertical FFC ZIF connector
• Power pins are 24V (2), 5V (2), 3.3V (2) and GND (11)

Peripheral controller (ARM7)
• Handles weigher, customer display & printer functions

• Weigher circuit is based on Analog Devices AD7730 delta sigma A/D converter
    • 5VDC analog supply drives load cell, differential amplifier & A/D front end
    • A/D conversion is ratiometric to load cell power
    • Diff amp has low pass filtering and gain of 9
    • Temperature stability & noise is critical to diff amp circuit
    • AD7730 has gain, calibration & digital filtering built in
    • Sampled at 30 samples per second
    • ARM7 performs final calibration, zero maintenance, filtering & motion detection

• ARM7 controls VFD customer display via 1 wire half duplex UART port
    • ARM7 handles marquee animation effects
    • VFD refreshes multiplexed display

• Labels imaged by LX800 system controller and sent in blocks to ARM7 via USB
    • 3 USB buffers loaded & used in a "round robin" method
• Printer "teach table" (primitive commands) allows modification of printer functions
• ARM7 controls printhead loads & timing
    • History control reduces energy to dots on in previous line
    • PWM strobe control maintains lower peak dot temperatures
    • Printhead strobe protection circuit
    • Printhead driven from fixed 24V supply (switched off when not printing)
    • Timing table adjustments for printhead resistance and temperature variations
• Power supply current monitored and printer timing adjusted to prevent power supply overloads with dense printing
• Head up/cassette type (1 of 3) sensor integrated with media sync sensor board
• Media sync sensor auto calibrates LED current and tracks detector voltage to dynamically adjust sync bar threshold for label stock & temperature variations
• Label taken sensor LED is driven at 250 mA with 1/8 duty cycle to be brighter than ambient light at retroreflective detector

Power supplies & sequencing

- 24 VDC input to main bd
- 5VDC synchronous switching supply (FAN5234) powered from 24VDC
- 3.3VDC switching supply (LM3475) powered from 5VDC

- System controller circuit – 3.3V I/O, 2.6V DDR400 & 1.25V Core
  - 2.6V & 1.25V are switching supplies (LM3475) powered from 5VDC
  - 2.6V & 1.25V supplies are enabled when 5V power is good
  - 3.3V supply is enabled after 2.6V powers up

- ARM7 – 3.3V I/O & 1.8V Core
  - 1.8VDC supply is low drop out linear regulator powered from 3.3VDC
  - 1.8VDC powers only the ARM7 core
  - ARM7 requires 1.8VDC to track below 3.3VDC during power up & power down, which is inherent to low dropout regulators

- Weigher – dedicated 5VDC linear, to ensure quiet A/D circuit

Reset button

- The reset pushbutton has a 0.5 to 1 second hardware delay to ensure that an inadvertent prees does not cause the HLX to go into shutdown
- A reset button press longer than the HW delay causes the ACPI circuit in the companion chip to perform an orderly scale application and Windows shutdown.
- A reset button hold for 4 seconds after the HW delay causes and immediate shutdown. Power is not removed from the main board (like in a PC) but is negligible.
- Pressing the reset pushbutton when in the shutdown state causes the BIOS to boot up Windows and the scale application.

Compact Flash & Hard Drive

- Either Compact FLASH or Hard Drive is used in an HLX (not both)



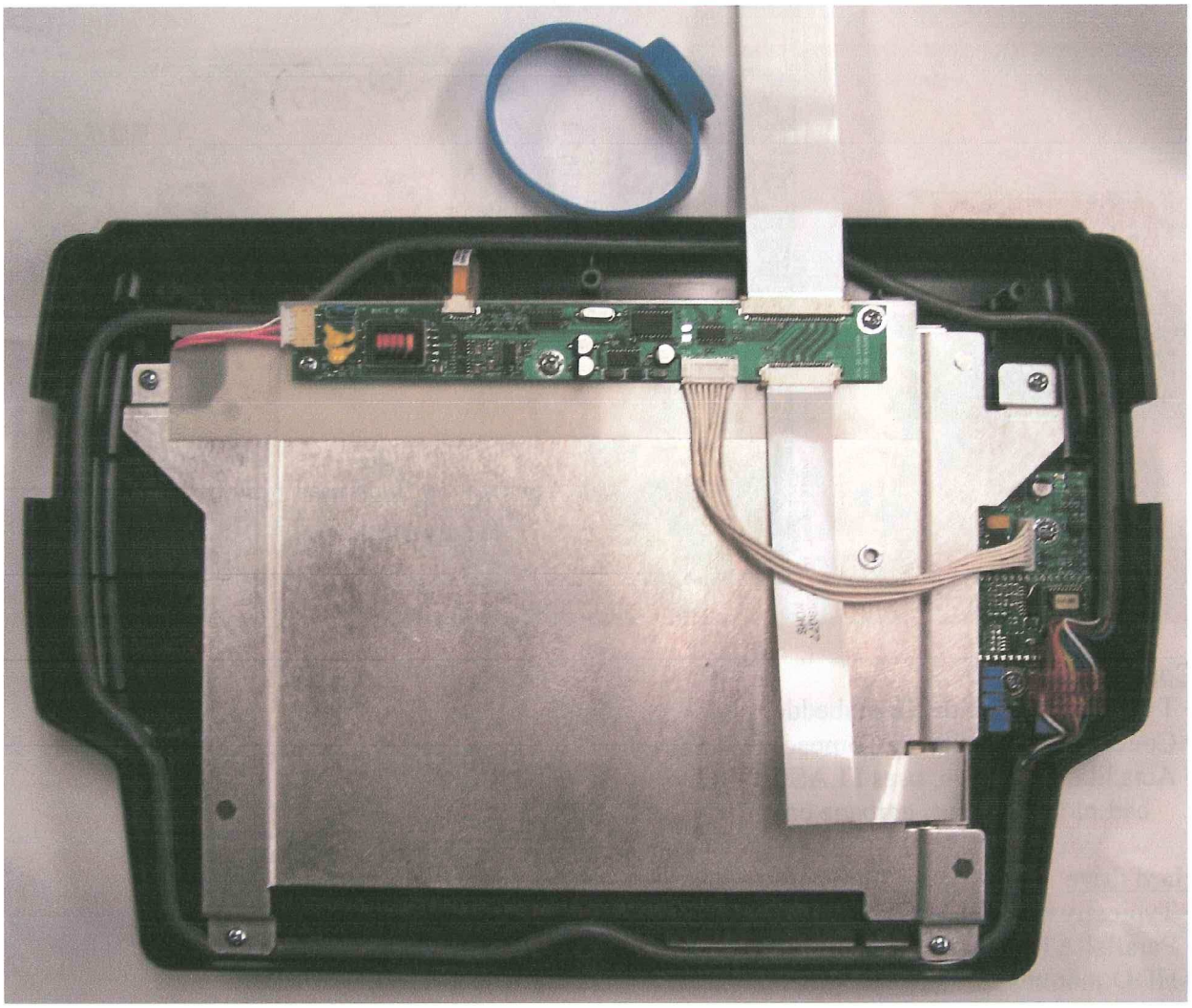
Compact FLASH

- Typically used with XP embedded
- Code and data reside in Compact FLASH
- Acts like disk drive, uses FLASH File System (FFS) in SW including, bad block lockout, garbage collection, ECC, FAT & wear leveling

Hard drive

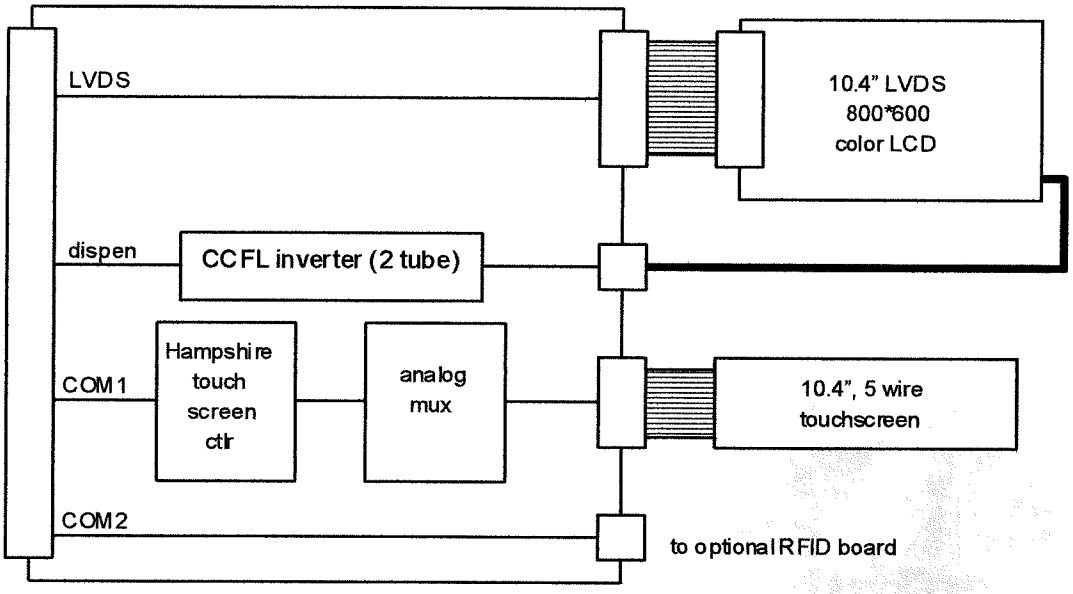
- Typically used with XP Pro
- Parallel ATA, notebook style (2.5")
- HDD mounted to HDD adapter bd, which is installed with 1 screw
- No HDD cable needed
- Another HDD adapter bd developed for serial ATA drives

HLX Console



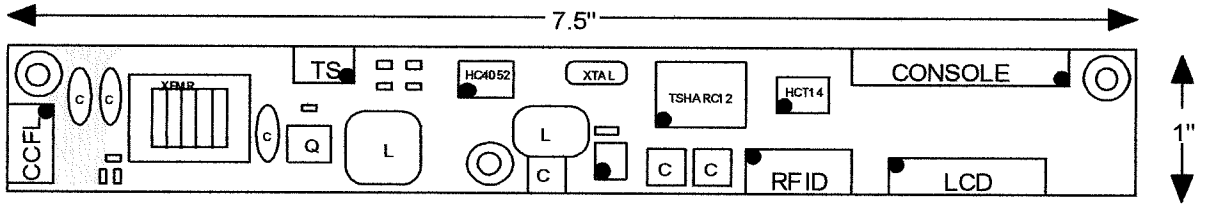
- LCD is 10.4", SVGA (800*600), LVDS interface
    - 2 CCFL tubes for ~400 nit brightness
    - 2 sources- AUO & Prime View
- LCD signals & power fed through console cable, console bd & LCD cable to LCD panel
- Touch screen is 5 wire style for longer life than 4 wire (35 vs. 1 million operations)
    - Standard glass with custom tail for minimum cost yet meets physical constraints
- COM1 drives Hampshire touch screen controller IC (PIC micro with special code)
    - Hampshire IC can control 4. 5 or 8 wire touch screens
    - Windows drivers and utilities furnished by Hampshire
- COM2 drives optional RFID board, used for operator sign on functions
- RFID board (far right) derived loop antenna along periphery of console housing
- Also shown is operator RFID wristband

HLX Console Board Block Diagram & Layout



30 pos 1mm
FFC connector

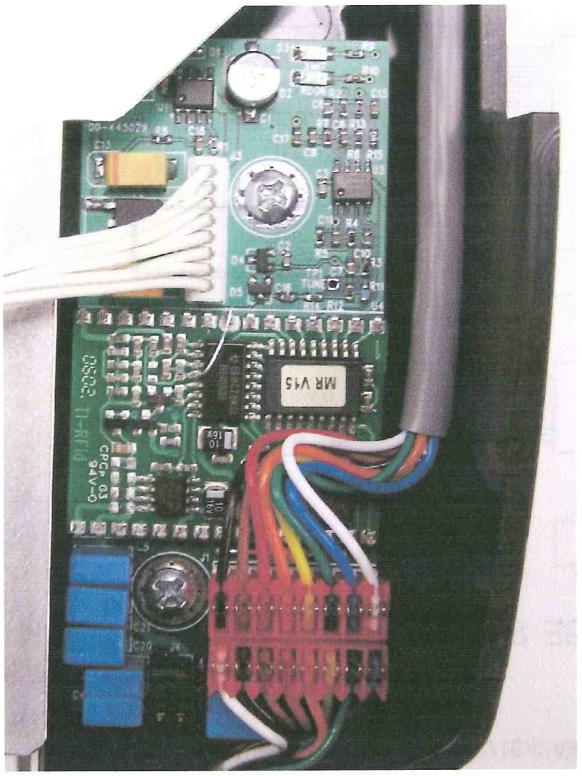
MIDRANGE & HIGH ENDCONSOLE BLOCK DIAGRAM
(with console parts shown)
1/17/2006 (rev 5/31/2006) , MEE, filename: CONMID.CDD

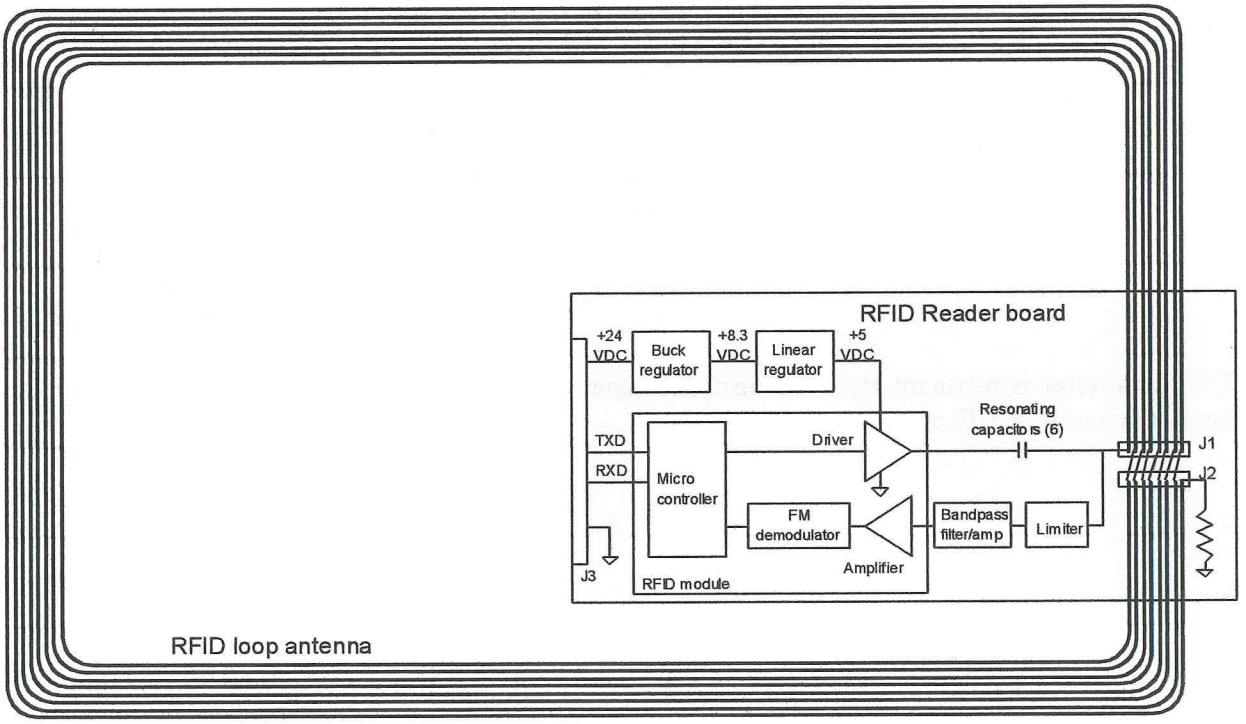


CONSOLE BD LAYOUT
7/31/2006 MEE
file: CON3.CDD

• CCFL inverter is resonant style for best efficiency and frequency locked to 80 KHz to reduce interference with RFID circuit

RFID option photo & block diagram

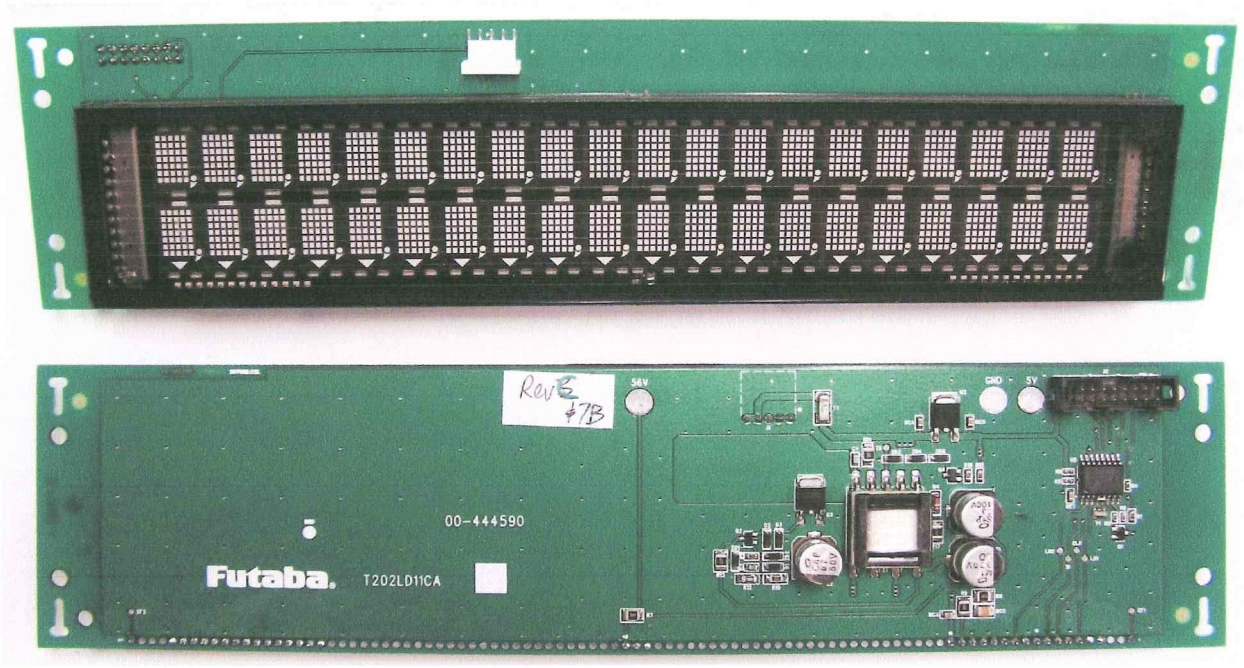


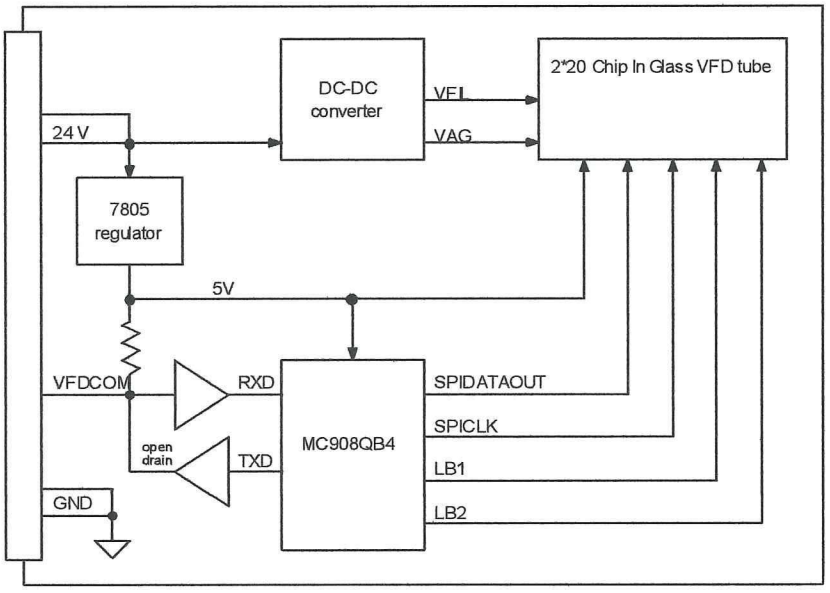


April 26, 2007

- Low Frequency RFID (134 KHZ) using TI RFID module & transponders
- Transmitter drives resonant circuit (resonating capacitors & loop antenna) with intentionally low Q to reduce external influences
- Magnetic field charges wristband transponder
- When transmission stops, transponder transmits 64 bit ID code, via FM at 122-134 KHz
- Hobart designed high gain, high Q bandpass filter allows 9"read range
- Many HLX circuits were tweaked to reduce Radio Frequency Interference in RFID receiver
- FCC certified as intentional radiator

Customer display



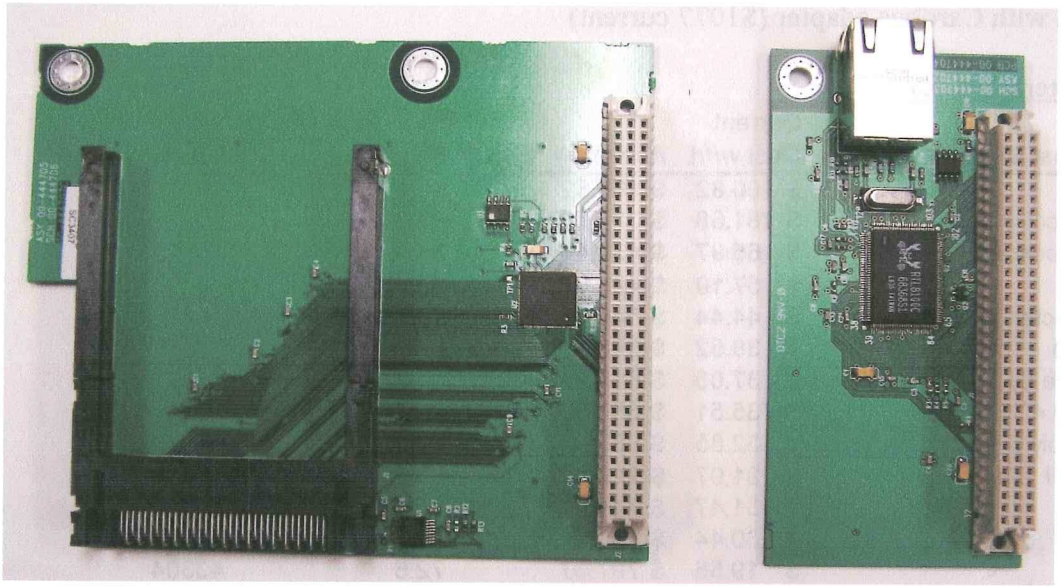


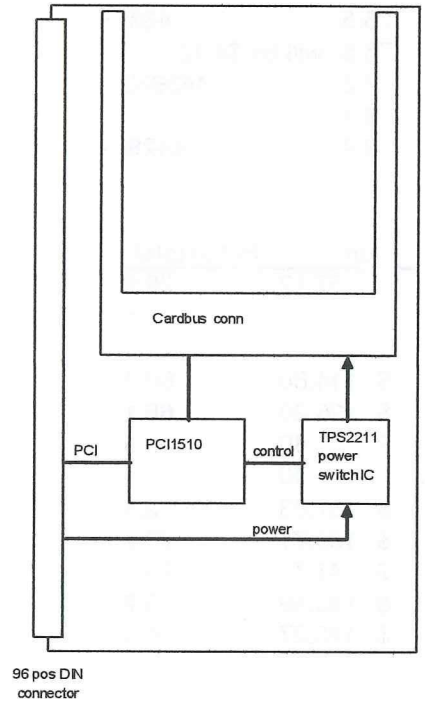
5 pos MTA-100 right angle connector

VFD BD BLOCK DIAGRAM
8/28/2006 , MEE, filename: VFD2.CDD

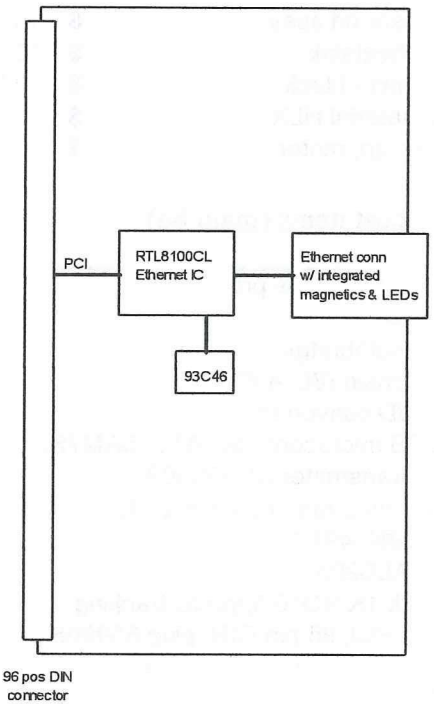
- 2 * 20 character vacuum fluorescent display with driver chips in glass
  - top line is animated marquee message
  - bottom line is weight & price declaration
- Futaba designed hardware & Futaba manufactured
- Hobart designed firmware and comm protocol

LAN Options





CARDBUS BD BLOCK DIAGRAM
rev 2/20/2006, MEE, filename: CARDBUS.CDD



ETHERNET LAN BD BLOCK DIAGRAM
rev 5/31/2006, MEE, filename: LAN.CDD

- Customer can choose one of two LAN options
- Cardbus board enables installation of 802.11 b/g RFLAN cards with Cardbus, PCMCIA & Compact FLASH interfaces (CF uses saddle adapter)
- Ethernet board allows 10/100 baseT Ethernet using Realtek RTL8100 IC

In warehouse cost (Frozen 11/11/2007)

- $1099 – HLX with Ethernet ($1068 current)
- $1107 – HLX with Cardbus adapter ($1077 current)

Top 80% cost items(HLX)

HLX top 80% cost items	Current Cost w/frt	Accum	% of IWHC	HLX with Ethernet $ 1,099
HLX main bd	$ 190.82	$ 190.82	17.4	
LCD 10.4" TFT color 800*600	$ 181.68	$ 372.50	33.9	
Windows XPe license	$ 65.97	$ 438.47	39.9	
Cassette assy	$ 57.10	$ 495.57	45.1	
VFD 2*20 character	$ 44.44	$ 540.01	49.1	
Printhead (TDK)	$ 38.52	$ 578.53	52.6	will be $30.96
SODIMM Module DDR400 512MB	$ 37.05	$ 615.58	56.0	will be $29.65
Touchscreen 10.4" 5 wire	$ 35.51	$ 651.09	59.2	445363
HLX power supply (Power One)	$ 32.85	$ 683.94	62.2	will be $24.35
Compact FLASH 1GB	$ 31.97	$ 715.91	65.1	
Loadcell	$ 31.47	$ 747.37	68.0	
Main base	$ 30.44	$ 777.81	70.8	444750
Platter assy	$ 19.56	$ 797.37	72.6	43304
HLX Ethernet bd assy	$ 16.32	$ 813.69	74.0	
Media sensor bd assy	$ 16.21	$ 829.90	75.5	43358
Printhead heatsink	$ 10.96	$ 840.86	76.5	will be $4.72
Main housing - black	$ 9.03	$ 849.90	77.3	443633-2
Packing material HLX	$ 8.50	$ 858.39	78.1	
Pulley take up, motor	$ 3.58	$ 861.97	78.4	442822

Top 80% cost items (main bd)

Total	$ 190.82	Sum	Pct of total
CM assembly labor & profit	$ 57.12	$ 57.12	29.9
LX800 CPU	$ 33.25	$ 90.37	47.4
CS5536 Southbridge	$ 12.35	$ 102.72	53.8
board, pc, main (8L, 4.4" *9")	$ 11.88	$ 114.60	60.1
AD7730 A/D converter	$ 11.60	$ 126.20	66.1
ARM7 USB microcontroller AT91SAM7S256	$ 6.60	$ 132.80	69.6
IC, LVDS transmitter DS90C363	$ 2.50	$ 135.30	70.9
A3983SLP step motor controller IC	$ 2.33	$ 137.63	72.1
IC, clock, MK1491-09	$ 2.08	$ 139.71	73.2
IC, AC97 ALC203	$ 2.00	$ 141.71	74.3
res network 1K SO16 5ppm/C tracking	$ 1.78	$ 143.49	75.2
conn, LAN PCI, 96 pin DIN, plug 650908-5	$ 1.78	$ 145.27	76.1
conn, IDE 50 pos 2mm shrouded	$ 1.78	$ 147.05	77.1
AD8672 dual op amp	$ 1.70	$ 148.75	78.0
IC, FWH ST M50FW040K5G	$ 1.68	$ 150.43	78.8
conn, console 30 pos FFC vert 1mm	$ 1.45	$ 151.89	79.6
conn, SODIMM 200 rvs tall Foxconn AS0A426BAR	$ 1.24	$ 153.12	80.2
cap, 3900/35 Panasonic FC series	$ 0.97	$ 154.09	80.8

Cost reduction plans/opportunities

- $11 – Weigher A/D circuit redesign
- $7 – Printhead alternate vendor
- $2 – DDRAM
- $8 – Power supply alternate vendor
- $6 – Printhead heat sink simplification
- $34 – Total

Section B - 00-444633 LX SCALE MAIN BOARD Schematic

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5536 / PCI Slot / LAN Slot	Page 8
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Board Stack-Up	Page 22

⊕ ADD ⊖ SUBTRACT ● WAS

LTR	REVISION	ZONE	DATE	CH NO.
B	RELEASED TO PRODUCTION		02-09-07	2022114
C	Changed R251 & R252 to 33.2 ohms Changed L4 & L5 to 6.8uH	5-C3 13-B6	04-12-07	2022163
	Changed pushbutton functionality to power button	13-B2		

IBIS Models used for simulations

LX800 - 33233_lx.ibs
5536 - 40144a_cs5536_b1.ibs
MK1491 - mk149_09.ibs
Memory - MT8VDDT3264HDG-40bF4.ebd

ISSUE LIST FROM REFERENCE DESIGN

DCC DS

DRAWING STATUS

STATUS

SUPERSEDED BY

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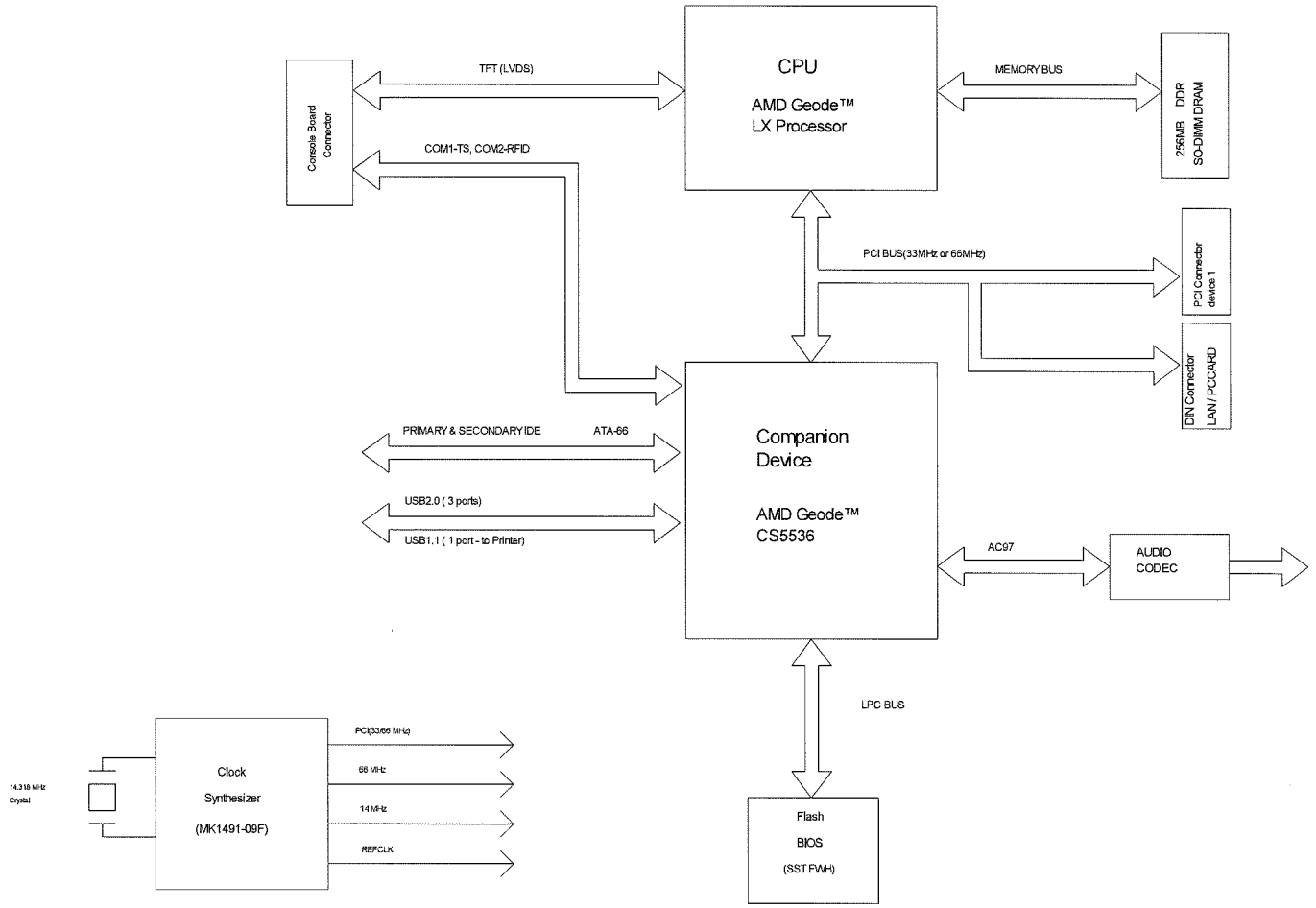
HOBART

NAME: SCH, LXSCALE MAIN BOARD

SUBCIRCUIT

VIEWDRAW PROJECT	HLX SCALE	FIRST USED ON	HLX SCALE
VIEWDRAW FILENAME	444635C		
REF. ASSEMBLY NUMBER	00-444635	REF. PCB NUMBER	00-444634
DWN BY	J. DUSING	DATE 04-10-07	LAST UPDATE 4-12-2007 10:55
CHK BY	M. EBERHARDT	DATE 04-11-07	DEV DWG

SIZE	DRAWING NO	SHEET	REV
C	00-444633	1 OF 22	C

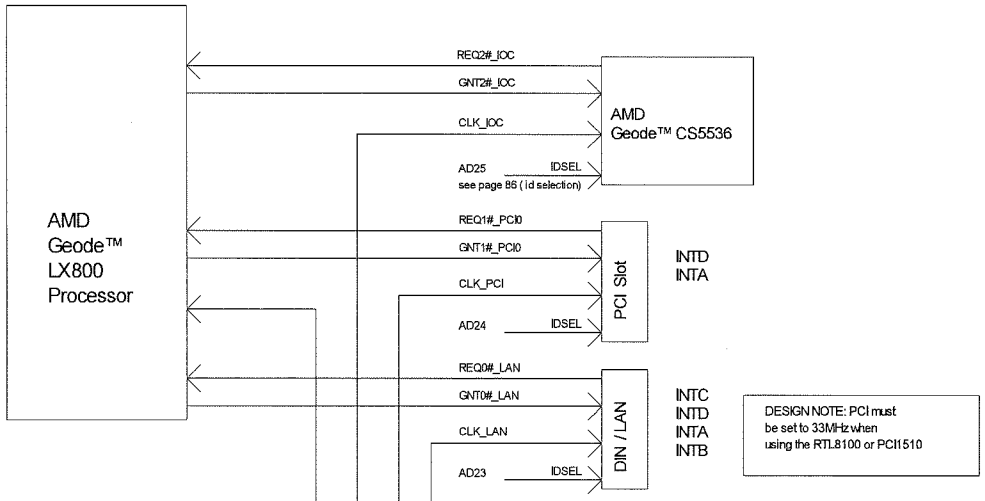


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HOBART			
NAME: SCH, LXSCALE MAIN BOARD			
DESCRIPTION: System Controller Block Diagram #1			
MENTOR PROJECT: HLX_SCALE		PARTITION: HLX_SCALE	
MENTOR RELEASE: 444635C		DOC: DS	
REF ASSEMBLY NUMBER: 00-444635		REF PCB NUMBER: 00-444634	
DRAWN: J. DUSING	DATE: 04-10-07	LAST UPDATE: 4-12-2007 10:55	
CHECKED: M. EBERHARDT	DATE: 04-11-07	REVISED:	
	SIZE: C	DRAWING NUMBER: 00-44463?	SHEET: 2 OF 22
			REV: C

LED , ,nts		
D7	USB Power Internal	Green
D8	+5V Power	Green
D9	Peripheral Error	Green
D13	Reset	Red
D17	USB Power External	Green
D19	IDE Activity	Green

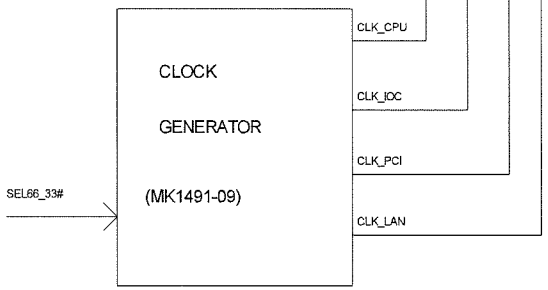
PCI MASTER SETTING



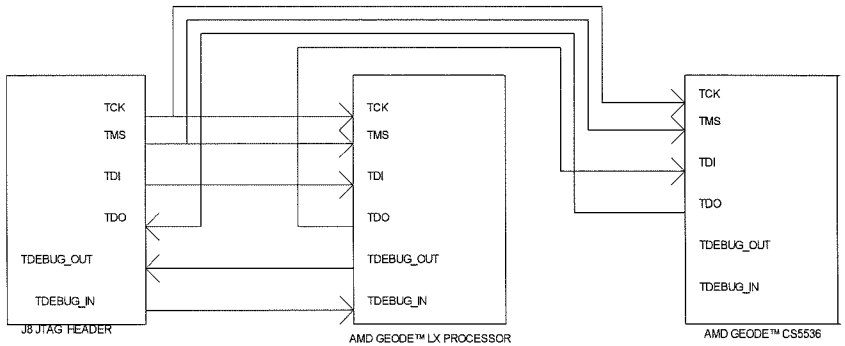
DESIGN NOTE: PCI must be set to 33MHz when using the RTL8100 or PCI1510

PCI CLOCK SETTING

SEL66_33#	Function
1	PCI 66 MHz
0	PCI 33 MHz

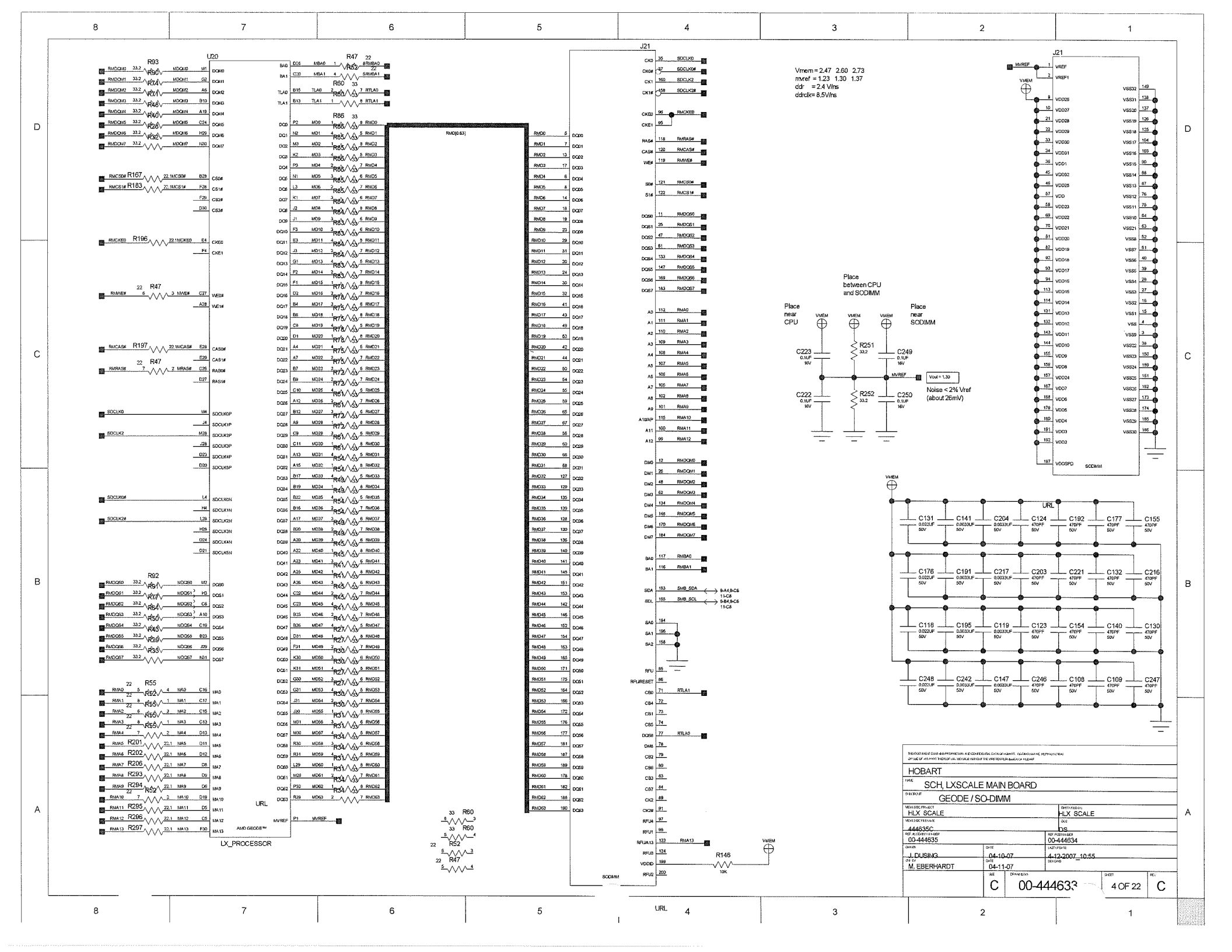


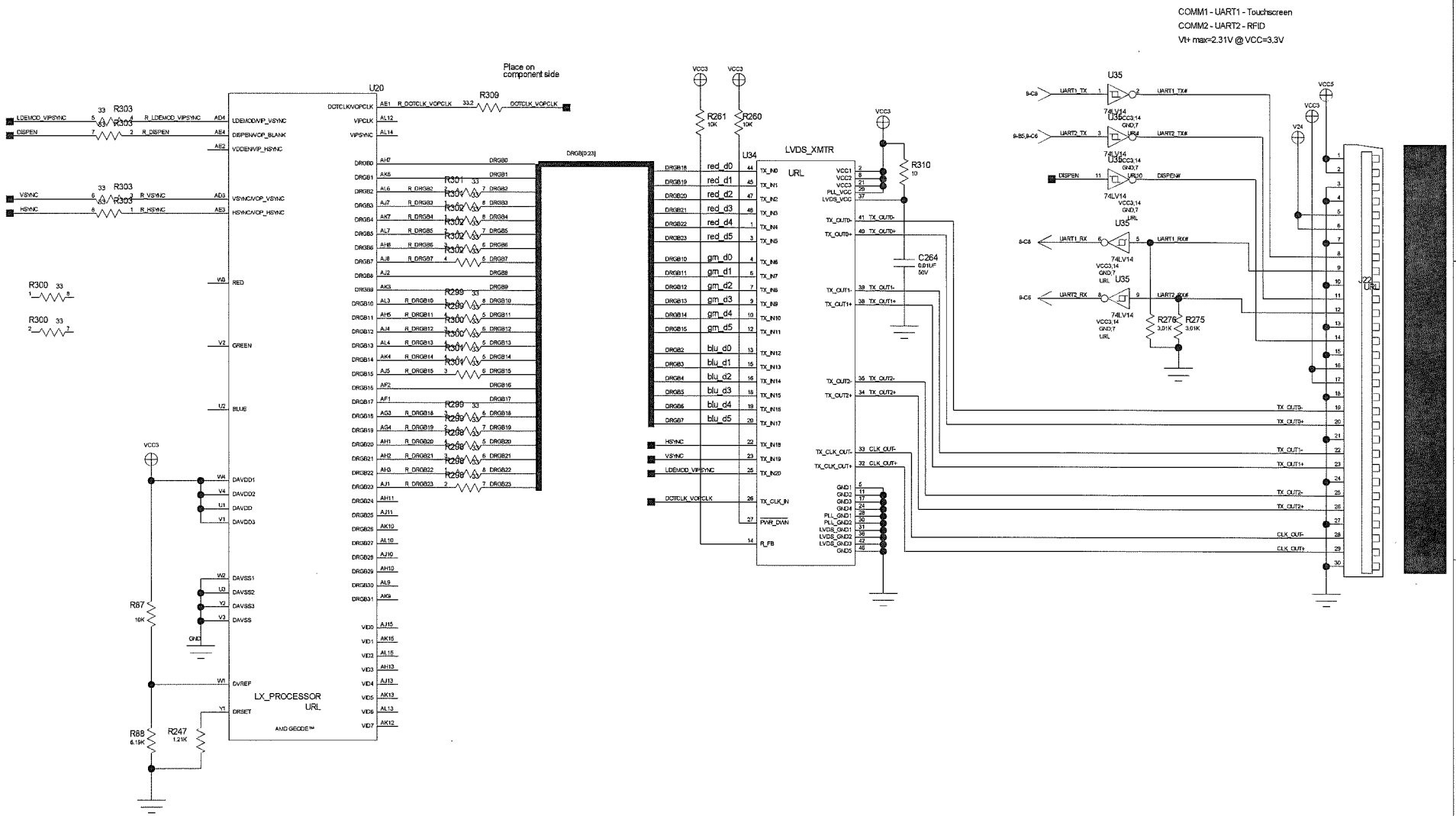
JTAG DAISY CHAIN MODE WITH AMD GEODE™ CS5536 COMPANION DEVICE



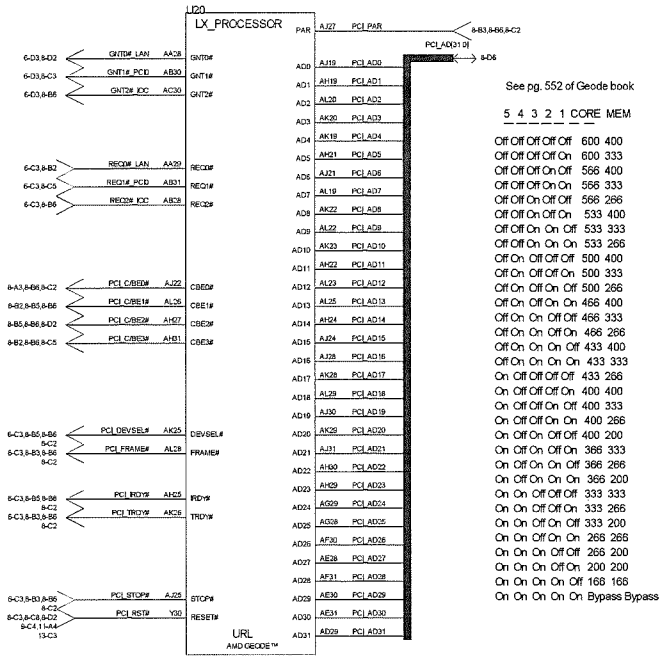
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NAME	HOBART
SCH, LXSCALE MAIN BOARD	
SHEET DESCRIPTION	System Controller Block Diagram #2
VIEW DOC PROJECT: HLX_SCALE	PREVIOUS ASSY: HLX_SCALE
VIEW DOC FILENAME: 444635C	DOC: DS
REF ALLOWED NUMBER: 00-444635	REF PCB NUMBER: 00-444634
DRAWN BY: J. DUSING — DATE: 04-10-07	LAST UPDATE: 4-12-2007_10:55
CHK'D BY: M. EBERHARDT — DATE: 04-11-07	REVISED
SIZE: C — DRAWING NO: 00-444633 — SHEET: 3 OF 22 — REV: C	

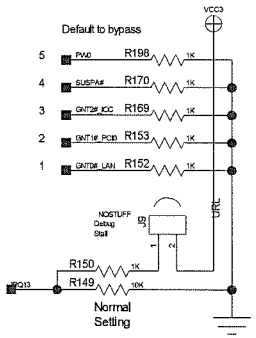

U20
LX_PROCESSOR
J21
SODIMM
Vmem = 2.47 2.60 2.73
mvref = 1.23 1.30 1.37
ddr = 2.4 V/ns
ddrclk= 8.5V/ns
Place between CPU and SODIMM
Place near CPU
Place near SODIMM
Vout = 1.30
Noise < 2% Vref (about 26mV)
HOBART
SCH, LXSCALE MAIN BOARD
GEODE / SO-DIMM
HLX SCALE
00-444635
00-444634
J. DUSING
04-10-07
M. EBERHARDT
04-11-07
C
00-444633
4 OF 22
C

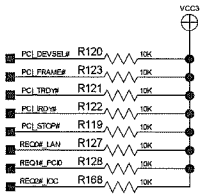


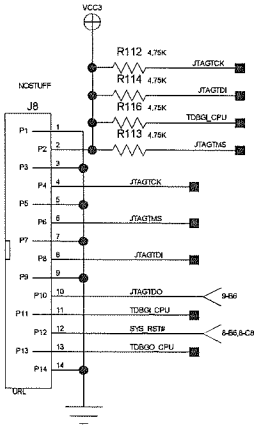
	HOBART
NAME	SCH, LXSCALE MAIN BOARD
SCHEMATIC	GEODE / Console Board Interface
VERILOGIC PROJECT	HLX_SCALE
PERFORMED ON	HLX_SCALE
VERILOGIC FILENAME	444635C
DAC	DS
REF NUMBER	00-444635
REF NUMBER	00-444634
DRAWN	J. DUSING
DATE	04-10-07
LAST MODIFIED	4-12-2007_10:55
CHECKED	M. EBERHARDT
DATE	04-11-07
DEV (VAL)	
SIZE	C
DRAWING NO	00-444633
SHEET	5 OF 22
REV	C

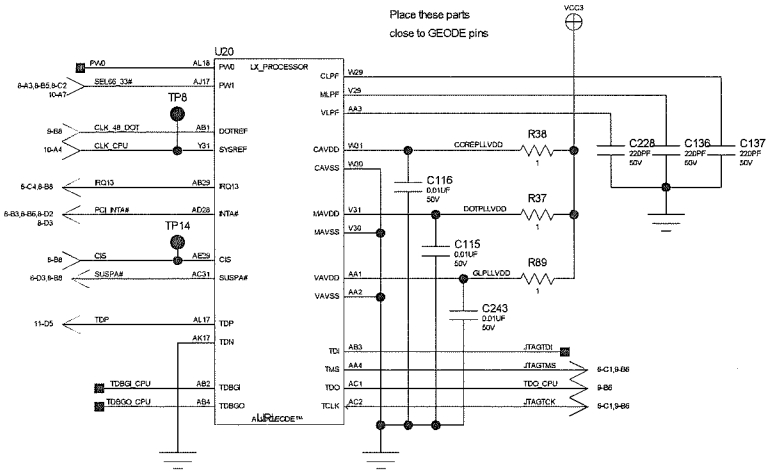


5	4	3	2	1	CORE	MEM
Off	Off	Off	Off	Off	600	400
Off	Off	Off	Off	On	600	333
Off	Off	Off	On	Off	566	400
Off	Off	Off	On	On	566	333
Off	Off	On	Off	Off	566	266
Off	Off	On	Off	On	533	400
Off	Off	On	On	Off	533	333
Off	Off	On	On	On	533	266
Off	On	Off	Off	Off	500	400
Off	On	Off	Off	On	500	333
Off	On	Off	On	Off	500	266
Off	On	Off	On	On	466	400
Off	On	On	Off	Off	466	333
Off	On	On	Off	On	466	266
Off	On	On	On	Off	433	400
Off	On	On	On	On	433	333
On	Off	Off	Off	Off	433	266
On	Off	Off	Off	On	400	400
On	Off	Off	On	Off	400	333
On	Off	Off	On	On	400	266
On	Off	On	Off	Off	400	200
On	Off	On	Off	On	366	333
On	Off	On	On	Off	366	266
On	Off	On	On	On	366	200
On	On	Off	Off	Off	333	333
On	On	Off	Off	On	333	266
On	On	Off	On	Off	333	200
On	On	Off	On	On	266	266
On	On	On	Off	Off	266	200
On	On	On	Off	On	200	200
On	On	On	On	Off	166	166
On	On	On	On	On	Bypass	Bypass









HOBART	
PAGE: SCH, LXSCALE MAIN BOARD	
SUBTITLE: GEODE PCI Bus	
XDW.IAC PROJECT: HLX_SCALE	PARTNUMBER: HLX_SCALE
XDW.IAC FILENAME: 444635C	DOC: DS
PCF ASSEMBLY NUMBER: 00-444635	PCF PARTNUMBER: 00-444634
DRAWN: J. DUSING	DATE: 04-10-07
CHKD: M. EBERHARDT	DATE: 04-11-07
	LASTSAVE: 4-12-2007_10:55
	DEVGRP:
SIZE: C	DRAWING NO: 00-44463?
SHEET: 6 OF 22	REV: C